

# Avantika A D

Aspiring Data Analyst — AI & Machine Learning Enthusiast

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**GitHub:** github.com/avantika-a-d — **Portfolio:** avantika-a-d.github.io

## Summary

Artificial Intelligence and Data Science student skilled in machine learning, data analytics, LLM security, and software development. Experienced in building predictive models, AI-powered applications, and adversarial testing tools using Python and modern ML frameworks. Passionate about solving real-world problems through AI, data, and cybersecurity with strong analytical and collaborative abilities.

## Education

**B.Tech – Artificial Intelligence and Data Science** 2023 – 2027  
SRM Valliammai Engineering College CGPA: 9.4

**Class XII** 2022 – 2023  
NSN Memorial Senior Secondary School 94%

## Skills

**Programming:** Python, SQL

**ML/AI:** Scikit-learn, XGBoost, TensorFlow, NLP, LLMs

**Data:** Pandas, NumPy, Matplotlib

**Web/Backend:** FastAPI, HTML, CSS, JS, REST APIs

**Security:** LLM Security, Adversarial Testing, OWASP LLM Top 10

**Tools:** Git, VS Code, Jupyter Notebook, Google Colab

## Projects

**Adversarial Testing Application** *Python, FastAPI, Groq LLaMA 3.3, REST API, NLP*

- Developed a red-teaming application to systematically test LLM-powered applications for security and safety vulnerabilities
- Automated adversarial prompt generation using Groq's LLaMA 3.3 API and classified vulnerabilities based on OWASP LLM Top 10
- Implemented asynchronous API testing, heuristic-based response scoring, and auditable evaluation workflows

**F1 Race Winner Prediction System** *Python, XGBoost, FastF1, Pandas, NumPy, Matplotlib*

- Developed and optimized machine learning model using FastF1 dataset (2021–2025 seasons)
- Performed feature engineering and data preprocessing on driver, constructor, and grid datasets
- Achieved 75–85% prediction accuracy using XGBoost

**ShieldBot – AI Safety Chatbot** *Python, Ollama, NLP, FastAPI, HTML/CSS/JS*

- Developed a chatbot to detect and handle risky user queries
- Deployed multi-level risk classification system
- Applied NLP techniques and risk classification models for safe AI response handling

**EcoLife – Sustainable Shopping Platform** *Python, OCR (Tesseract), ML, Google Shopping API, React*

- Created AI-powered platform using OCR (Tesseract) for product analysis
- Generated eco-scores and alternative recommendations

## Experience

**eMudhra Limited – Technology Innovation Intern (Onsite - Bangalore)** Jun 2026 – Jul 2026

- Developed an adversarial testing application to red-team Lia, an LLM-powered digital signing and certificate assistant
- Automated adversarial prompt generation using Groq's LLaMA 3.3 API to identify security and safety weaknesses
- Implemented REST API integration, asynchronous testing, heuristic-based response scoring, and vulnerability classification
- Applied OWASP Top 10 for Large Language Model Applications to structure security testing and evaluation workflows

**TATA Forage – Data Analytics Micro Internship (Remote)** Jan 2025 – Feb 2025

- Analyzed and interpreted datasets containing 200+ records to identify business trends and actionable insights
- Created 5+ data visualizations and dashboards to support data-driven decision-making
- Improved reporting efficiency by applying structured data analysis techniques using Python and Excel

## Relevant Coursework

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Machine Learning, Data Analytics, Big Data Analytics, Cloud Computing, Text & Speech Analysis, Database Management Systems, Data Structures and Algorithms, Artificial Intelligence

## Certifications

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Generative AI Foundations – AWS

Data Visualization: Empowering Business with Effective Insights – Tata Forage

Perl – IIT Bombay FOSS

AI and its Applications – SRM Valliammai Engineering College

## Achievements & Leadership

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- Secured first rank in 3 out of 5 semesters in B.Tech (AI and DS)
- Served as **Secretary** for the Department Symposium “ALTRUIXX’26”, coordinating office bearers and overseeing the planning and execution of the symposium
- Organized and coordinated the technical event “Comic Clash” at the college
- Participated in hackathons and inter-college technical competitions

## Soft Skills

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Analytical Thinking, Problem-Solving, Team Collaboration, Adaptability, Effective Communication, Attention to Detail